LIL: Lightweight Incremental Learning Approach Through Feature Transfer for Remote Sensing Image Scene Classification

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***Abstract*— Existing deep learning models usually assume that all data obeys independent identically distribution, which is unreasonable in remote sensing. Due to the differences in camera parameters, spectral ranges, resolutions, and so on, the images acquired by remote sensing sensors may be greatly diverse, causing models to face catastrophic forgetting when they are trained on new data only. Thus, incremental learning is introduced. An ideal incremental learning model should be expanded as the number of tasks increases, so as to have enough ability to adapt to the changes in data. However, existing approaches normally expand heavy modules for each task, making the holistic models cumbersome. In this article, a lightweight incremental learning approach (LIL) is proposed for remote sensing image scene classification. We replace the role of the feature extractor with extracting features of a single task instead of task-sharing features of all tasks to lighten the backbone. In addition, we propose a light feature transfer module (FTM) to realize the alignment of data distributions between different tasks in the feature domain. Furthermore, dual-constraint loss with knowledge distillation and adversarial learning is introduced to promote the mapping and alignment of data distributions at both the feature level and the semantic level. In LIL, only a tiny FTM and a classifier are added to the model when the model learns a new task. Experimental results show that our approach with a small number of parameters outperforms state-of-the-art approaches for incremental learning on both a single dataset and a sequence of multiple datasets.**

***Index Terms*— Feature transfer, image scene classification, incremental learning, lightweight model, remote sensing.**

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I. Introduction

MOTIVATED by deep neural networks (DNNs), remote sensing image interpretation has made great progress in recent years. It plays an important role in earth observation, disaster monitoring, urban planning, safety prevention, and so on. Because of the differences in shooting areas, weather conditions, complex geometrical structures, and spatial patterns, the images captured by remote sensing platforms vary mightily. However, existing deep learning models are usually designed and trained for the same scenario and do not have the ability to adapt to changes automatically. On the one hand, the inference of directly applying models to new data (new task) will immensely reduce the performance of models if newly obtained data are quite different from previous data (old task). On the other hand, the models will forget the knowledge of processing old data after fine-tuning (FT) on new data, known as catastrophic forgetting [1], [2]. An example of catastrophic forgetting is shown in Fig. 1.

The main reason for catastrophic forgetting is the differences in distributions between new data and old data. As shown in Fig. 1, the distributions of images in two training sets are awfully different. If the newly obtained data and the previously trained data obey diverse distributions, the models will face catastrophic forgetting after FT on the new data. As is well known, common deep learning models generally follow an idealized assumption that all data during training and testing obeys independent identically distribution (i.i.d.). It is inauthentic in real applications, especially in the field of remote sensing. Unlike natural images usually produced by complementary metal–oxide–semiconductor (CMOS) sensors, remote sensing images are acquired by sensors with different loads, resolutions, and spectral ranges. Therefore, the changes in remote sensing images are very complicated, and the distributions of remote sensing images are extremely different. In such a case, incremental learning [3], [4] is proposed to alleviate catastrophic forgetting. It expects models to have the ability to continuously learn new tasks while maintaining the performance of old tasks as much as possible so that the models can automatically adapt to different distributions of data. As a technology with practical application prospects,

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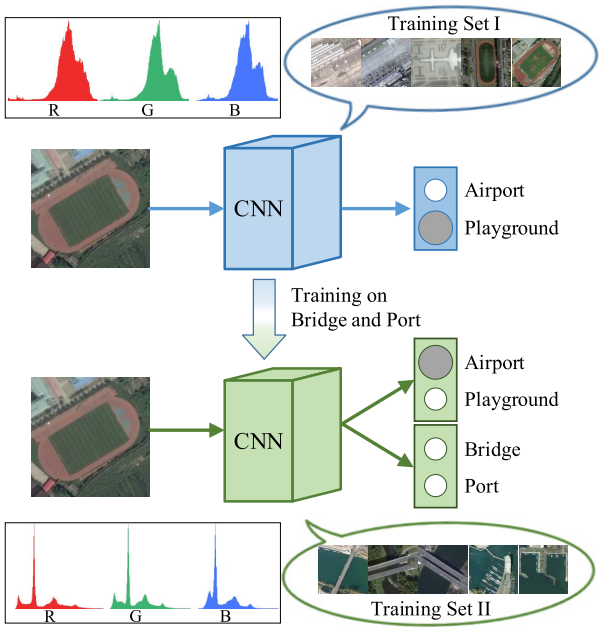


Fig. 1. Illustration of catastrophic forgetting. (Top) Original model, which is trained on Training Set I containing images of *Airport* and *Playground*, can correctly classify the input image of *Playground*. (Bottom) Updated model is fine-tuned on Training Set II, which only contains new images of *Bridge* and *Port* based on the original model. It mistakenly predicts the *Playground* image as an *Airport*, resulting in catastrophic forgetting.

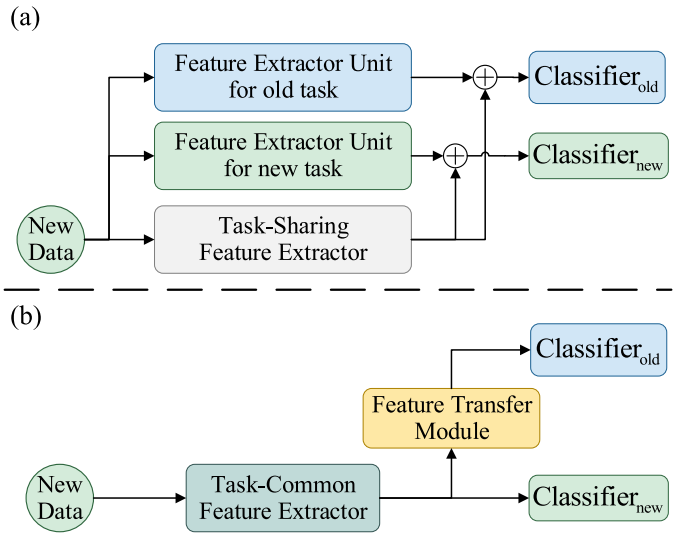


Fig. 2. Schematic of parameter expansion-based incremental learning methods. (a) Previous work has a large-capacity task-sharing feature extractor and expands heavy task-specific modules for each task. (b) TCE in our method is tiny, and only a light FTM and a classifier are added for learning a new task.

incremental learning will become one of the most valuable research directions in remote sensing.

According to the way of storing and using of specific task information and implementation strategies [4], incremental learning methods are divided into three families: replay-based methods [5]–[15], regularization-based methods [16]–[32], and parameter expansion-based methods [33]–[44].

When learning a new task, replay-based methods preserve old data or train generators to create pseudodata for joint training (JT) with new data, and regularization-based methods add regularization items to constrain the changes of important parameters for old tasks. These two branches of incremental learning approaches only learn tasks on the initially established models and do not expand networks from beginning to end. The capacities of models are fixed after initial constructions, which is not conducive for incremental learning. As the number of learning tasks increases, the parameters of models that can be used for learning new tasks become less. The models either forget the old knowledge learned before or fail to learn new tasks. Especially in the field of remote sensing, the images taken by satellites in multifarious scenarios correspond to various distributions, representing diverse tasks. In order to reach glorious incremental learning performance on remote sensing images, the architecture of an ideal incremental learning model is supposed to be expanded with the increase in the number of learning tasks. Thus, parameter expansion-based methods are the most suitable ways for incremental learning of remote sensing images.

However, the existing parameter expansion-based incremental learning methods generally add bulky modules for learning new tasks, as shown in Fig. 2(a). This leads to a heavy burden on the overall network, which is adverse to the applications on remote sensing platforms. In the field of remote sensing, the deployed models are anticipated to be as compact as possible. Therefore, it is necessary to carry out a lightweight design for parameter expansion-based incremental learning models of remote sensing.

There are two main reasons why existing parameter expansion-based models are cumbersome.

A. Task-Sharing Feature Extractor Large-Capacity Issue

The feature extractors of parameter expansion-based methods are generally introduced as task-sharing modules to extract mutual sharing features from images of all tasks. Their architectures need to be determined at the first stage of model construction. Thus, it is necessary to apply a large-capacity backbone for the sake of the learning of all tasks. In addition, the scenes in remote sensing images are vast and complex, and contain diversified information. Extracting shared knowledge from intricate and changeable information also requires a large-capacity network. More importantly, the shared features among tasks inevitably decrease as the number of tasks increases, resulting in the failure to mine numerous effective features. The large-capacity task-sharing feature extractor only learns a small amount of knowledge so that its capacity is squandered.

B. Task-Specific Module Redundancy Issue

The task-specific modules are usually added to parameter expansion-based models starting from the image domain. It requires that each module contains an independent unit for feature extraction, which increases the parameters of modules. Since the private feature extraction units and the task-sharing feature extractor learn features from the same images, there

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may be redundancies between them. Besides, remote sensing images have wide coverage and strong correlations between neighboring pixels, but there is no information interaction between task-specific modules. It is likely for respective modules of different tasks to learn the same knowledge, which gives rise to a substantial useless increase in the parameters of task-specific modules.

In view of the above two issues, our solutions for realizing a lightweight incremental learning (LIL) model are given as follows.

Aiming at the task-sharing feature extractor large-capacity issue, we no longer define the mission of the feature extractor as extracting sharing features among all tasks but for extracting features in the feature space of the last learned task. The feature extractor is still communal for all tasks, which we called task-common feature extractor (TCE). It merely learns to extract features of a specific task and does not desire a cumbersome backbone.

For the task-specific module redundancy issue, we propose a feature transfer module (FTM). Unlike task-specific modules of existing methods that process in the image domain directly and separately, FTM acts on the domain of features to learn the mapping functions of data distributions between adjacent tasks. It realizes the transformation of features from a new task to a contiguous old task. Since FTM only processes semantic features instead of images, it does not contain an independent feature extractor unit and can be designed to be light. Thus, feature transfer can be achieved with a few convolutional layers. Meanwhile, in addition to the routine distillation loss, inspired by the generative adversarial network (GAN) [45], we also introduce an adversarial loss, which together with the distillation loss constitutes a dual-constraint loss to supervise the training of the FTM at both the feature level and the semantic level.

Considering the above viewpoints, we propose an LIL framework. As shown in Fig. 2(b), our method only expands a tiny FTM and a classifier to learn a new task. For the ten-split aerial image dataset (AID), among parameter expansion-based methods, the architecture of our method is the lightest, and the number of parameters increases the slowest as the number of tasks increases. Besides, compared with the advanced parameter expansion-based method, ACL [44], our proposed LIL improves the mean accuracy (mACC) of all tasks by 4.29% and reduces the forgetting rate by 2.52%, while the number of parameters is decreased by about 74% (27.4 MB) of ACL without replaying the data of old tasks. Furthermore, our method without data replay also outperforms ACL with storing old data. Extensive experiments illustrate that LIL achieves state-of-the-art incremental learning performance with a small number of parameters on a single dataset and a sequence of multiple datasets.

Our contributions are summarized as follows.

1) We introduce an LIL model in the field of remote sensing image scene classification. It expands a few parameters as the number of learning tasks increases. Only a light FTM and a classifier are established for learning a new task. LIL addresses a lightweight design of incremental learning models and achieves state-of-the-art incremental learning performance with a tiny and handy architecture.
2) A new TCE is proposed to ease the backbone of the incremental learning model. We replace the role of the feature extractor to output features of the latest task only instead of extracting task-sharing features. The proposed TCE can be built of a tiny backbone because it merely extracts features of a single task.
3) We propose an FTM to transform features from a new task to the neighboring old task in the domain of features instead of images. Since the FTM just handles high-level features, it can be constructed to be light.
4) Dual-constraint loss is designed to manage FTMs. In addition to the conventional distillation loss applied at the feature level, we also present an adversarial loss based on adversarial learning to promote feature transfer at the semantic level. Dual-constraint loss promotes the training of FTMs efficiently.

The rest of this article is organized as follows. Section II briefly introduces three branches of incremental learning methods. Section III describes the general workflow and three key components of our proposed method. Sufficient experimental results and comparative analysis are provided in Section IV. Finally, conclusions are summarized in Section V.

II. Related Works

Driven by the widespread applications of deep learning, image scene classification and incremental learning have attracted more attention from researchers, and many works have been proposed. In this section, some related works of remote sensing image scene classification and three families of incremental learning methods are briefly introduced.

A. Remote Sensing Image Scene Classification

Before deep learning was widely developed, remote sensing image scene classification generally relied on artificially designed features [46]–[51]. With the development of remote sensing technology, a large number of high-resolution remote sensing images can be easily obtained. Thus, in the past few years, several remote sensing image scene classification datasets have been published, including visible spectral remote sensing datasets (AID [52], RS_C11 [53], RSSCN7 [54], SIRI-WHU [55], and SAT-4 [56]), multispectral datasets (EuroSAT [57] and BigEarthNet [58]), and synthetic aperture radar (SAR) datasets (MSTAR [59], TenGeoP-SARwv [60], and OpenSARUrban [61]). Propelled by available datasets and the powerful feature learning capabilities of DNNs, remote sensing image scene classification driven by deep learning has drawn remarkable attention and achieved significant breakthroughs [62]–[67]. Liu *et al.* [68] combine convolutional neural networks (CNNs) with hierarchical Wasseratein loss to improve the discriminative ability of models. Hydra [69] applies an ensemble of CNNs for land classification. Sumbul and Demir [70] propose a multiattention model to exploits CNNs and recurrent neural networks (RNNs) for multilabel classification. Fang *et al.* [71] propose a frequency domain

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branch for space-frequency joint learning. Byju *et al.* [72] propose a method for remote sensing image scene classification in the compressed domain. Kang and Demir [73] decouple the spectral and spatial feature extractions and introduce multiscale representations of different classes present in images. Li *et al.* [74] propose a cross-layer nonlocal (CLNL) framework to obtain the global contextual understanding ability.

These methods are proposed for traditional remote sensing image scene classification. The models only learn one classification task, and all data need to be available during training. These methods will face catastrophic forgetting if they are applied to incremental learning. Our proposed LIL can alleviate catastrophic forgetting while learning multiple tasks, in turn, without looking back at the data of old tasks.

B. Incremental Learning

1) Replay-Based Methods: Replay-based methods refer to storing a part of previously learned data in a specific form or training additional generative models to create pseudodata of old tasks. When models learn a new task, these old samples/pseudosamples are replayed for JT with new data. iCaRL [5], for each category, selects m samples closest to the average vector of the category and stores them to replay. Lopez-Paz and Ranzato [6] propose an episodic memory module to save old data. Riemer *et al.* [7] combine data replay with metalearning to balance the relationship between new and old tasks. In order to make models memory-friendly, Rannen *et al.* [11] and Iscen *et al.* [12] store the extracted features instead of input images. Shin *et al.* [8] train a generator to generate pseudosamples for JT. Memory replay GANs (MeRGANs) [9] and exemplar-supported generative reproduction (ESGR) [10] improve generators to generate preferable samples. In the field of remote sensing, Burzzone *et al.* [13] propose a radial-basis function (RBF) network for remote sensing image classification. Dang *et al.* [14] introduce an open set recognition algorithm to recognize new unknown classes continually. Li *et al.* [15] introduce a benchmark for incremental learning. CLDR [75] and PSGANs [76] apply GANs and data regeneration for remote sensing classification.

Replay-based methods call for additional memory to preserve old data or pseudodata generators. However, privacy concerns and memory constraints are important considerations in remote sensing. Compared with these works, LIL does not require extra memory to conserve old samples or extra models.

2) Regularization-Based Methods: Without the requirements for additional storage spaces, regularization-based methods apply knowledge distillation [77]–[80] or calculate the importance of parameters in some specific ways and then add one or more regularization items in loss functions to penalize the forgetting of the previous knowledge. Learning without forgetting (LWF) [16] and ExpertCNN [17] use the outputs of the old model as soft labels for distillation. Castro *et al.* [18] and Dhar *et al.* [19] fulfill end-to-end incremental learning. Elastic weight consolidation (EWC) [20] and its variants [21]–[23] calculate the importance of parameters and focus on protecting important parameters for old tasks from being changed. Ritter *et al.* [24] combine Bayesian online learning and Laplace approximation as a secondary penalty item. For object detection and semantic segmentation, some works [25]–[27] apply knowledge distillation to make the new model simulate the results of the old model on previous tasks. For remote sensing, Vetrivel *et al.* [28] realize incremental learning for disaster damage. Ammour *et al.* [29] use shared layers followed by task-specific layers to learn tasks continuously and introduce a task selector to determine the task to which the input image belongs. Tasar *et al.* [30] propose a method for semantic segmentation based on knowledge distillation. Feng *et al.* [81] propose pixel affinity structure loss and representation consistency structure loss for semantic segmentation continual learning. Dang *et al.* [31] use nonnegative matrix factorization to recognize targets in SAR images. Chen *et al.* [32] propose a method based on feature pyramid and knowledge distillation for incremental learning of remote sensing object detection.

Even if no storage space is needed, regularization-based methods may forget old knowledge due to insufficient punishments. They are powerless for model expansion that is required for incremental learning. On the contrary, the capacity of our proposed model increases slightly with the arrival of new tasks, which is conducive to incremental learning.

3) Parameter Expansion-Based Methods: In parameter expansion-based methods, different subparts of models are used for different tasks, or new branches are added for new tasks. Piggyback [33] fixes the parameters of a pretrained model and learns a mask to select the best part of parameters for each task. Packnet [34] releases redundant parameters after the training of each task through pruning and model retraining to free up the capacity of the model. Serra *et al.* [35] generate masks according to attention mechanism. Expert Gate [36] learns an autoencoder gate to free up the spaces for storing masks. Deep adaptation networks (DANs) [37] define new filters for learning a new task to be linear combinations of existing ones. Li *et al.* [38] use neural architecture search (NAS) [82], [83] for structural optimization to release parts of the model. There are also several works that extend separate parameters for new tasks. Reinforced continual learning (RCL) [39] expands the network through a controller, which decides how many nodes are added to each layer. Dynamically expandable networks (DENs) [40] and dynamic generative memory (DGM) [41] dynamically expand networks when learning new tasks. The progressive neural network (PNN) [42] explicitly structures the transformation between different tasks. It constructs a single-column network for each task and uses the knowledge of old tasks to promote the learning of new tasks through horizontal connections. PathNet [43] is an ultra-large-scale network. Unlike PNN, it directly prebuilds a modular neural network with M modules in each layer, and each module is an independent network. PathNet selects the best N modules in each layer when learning a new task. Adversarial continual learning (ACL) [44] factorizes features into shared and private features and builds an independent task-specific module for each task starting at the domain of images. CILEA-NET [84] introduces curriculum learning for remote sensing image classification.

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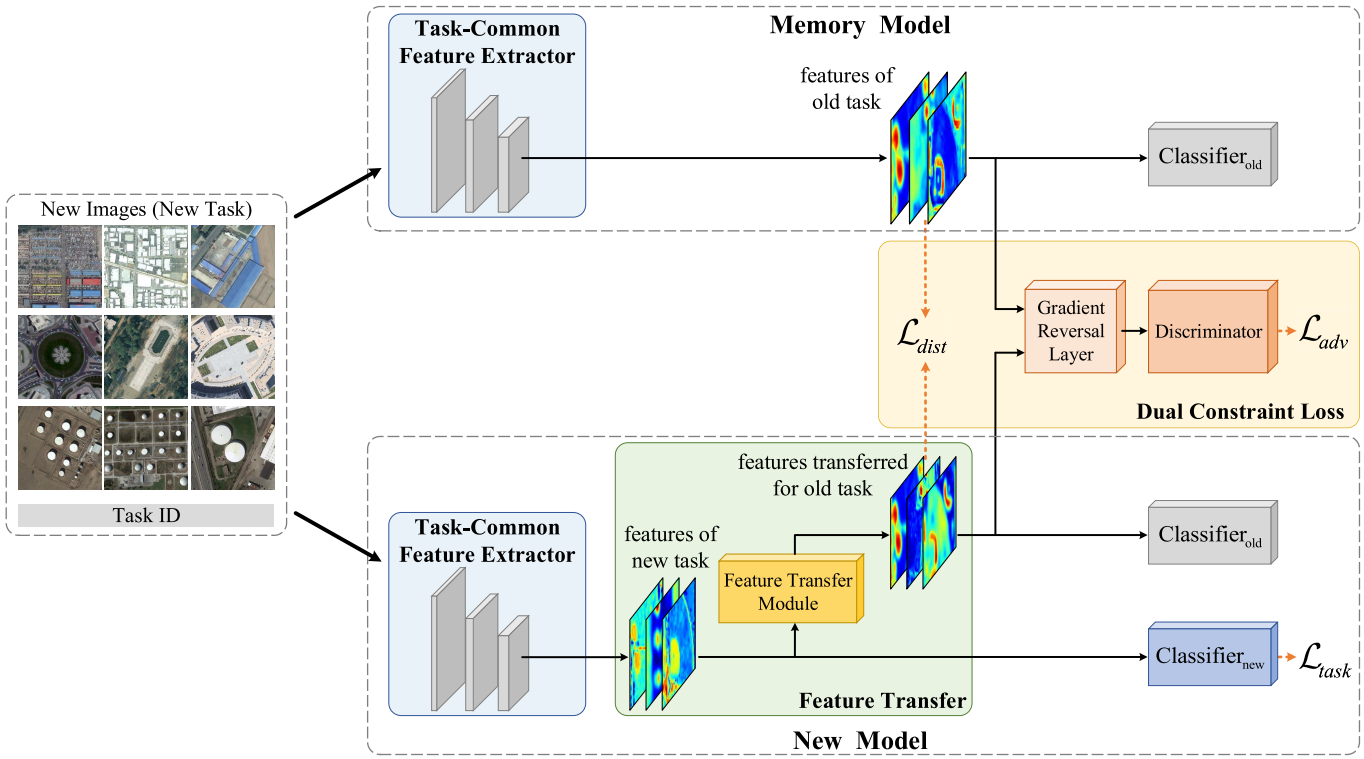


Fig. 3. Architecture of LIL. It includes TCE, FTM, and dual-constraint loss. (Top) When learning a new task, the model that performed well on old tasks is copied and froze as a memory model. (Bottom) Then, an FTM and a classification module (Classifier$_{new}$) are added to the original model to obtain a new model. A discriminator is initialized for adversarial learning in the dual-constraint loss. The classification loss of new task $\mathcal{L}_{task}$ and the dual-constraint loss ($\mathcal{L}_{adv}$ and $\mathcal{L}_{dist}$) are used together for the learning of the new model to consolidate old knowledge while learning the new task.

Although this kind of method increases the capacities of models, they expand relatively large modules that lead to a heavy burden on the overall models. Our method performs data alignment in the feature domain and only adds a lightweight FTM and a classification module to learn a new task, which makes the model light and handy.

III. Methodology

The details of our proposed method LIL are elaborated in this section. We first briefly introduce the overall architecture and the incremental learning workflow of LIL. Then, we separately present the components designed in LIL, including TCE, FTM, and dual-constraint loss.

A. Overview

As shown in Fig. 3, the proposed LIL mainly contains three parts: TCE, FTM, and dual-constraint loss. CNNs are used as the TCE to extract features from the input images in the feature space of the newest learned task, and the feature maps output by it are communal among all tasks. Furthermore, FTMs are built between adjacent tasks (according to the order of learning). The role of the FTM is to map features from a new task to the adjacent old task. The dual-constraint loss consisting of distillation loss and adversarial loss is introduced to supervise the learning of FTMs. Same as conventional classification models, both TCE and FTM are followed by independent task-specific classifiers that predict categories based on the extracted and transformed features.

The process of learning a new task in LIL is briefly introduced as follows. Assuming that the model performs well on the old task and needs to learn a new task, as shown in Fig. 3, we copy the old model and fix its parameters as the memory model (top) to guide the learning of the new model (bottom). The new model is built by adding an FTM and a classifier for the new task on the basis of the old model.

During training, the classification loss $\mathcal{L}_{task}$ is used to learn the new task. The dual-constraint loss is proposed to train the newly added FTM in both feature level and semantic level. It contains distillation loss $\mathcal{L}_{dist}$ and adversarial loss $\mathcal{L}_{adv}$. At the feature level, $\mathcal{L}_{dist}$ is used between the features extracted by the TCE of the memory model and converted by the FTM of the new model for training the FTM to convert features in the feature space from the new task to the old task.

In addition, a discriminator and an adversarial loss $\mathcal{L}_{adv}$ are constructed to supervise the learning of the FTM at the semantic level. Together with the FTM and the TCE, the discriminator forms an adversarial learning framework. The features extracted by the TCE of the memory model and transformed by the newly added FTM of the new model are input to the discriminator. The discriminator predicts whether the input features are extracted from the memory model or converted by the FTM of the new model. In the adversarial training framework, the discriminator learns to distinguish these two kinds of features as much as possible, while the FTM is trained to obfuscate the discriminator so that it cannot trace the source of the input features. In this way, the FTM learns

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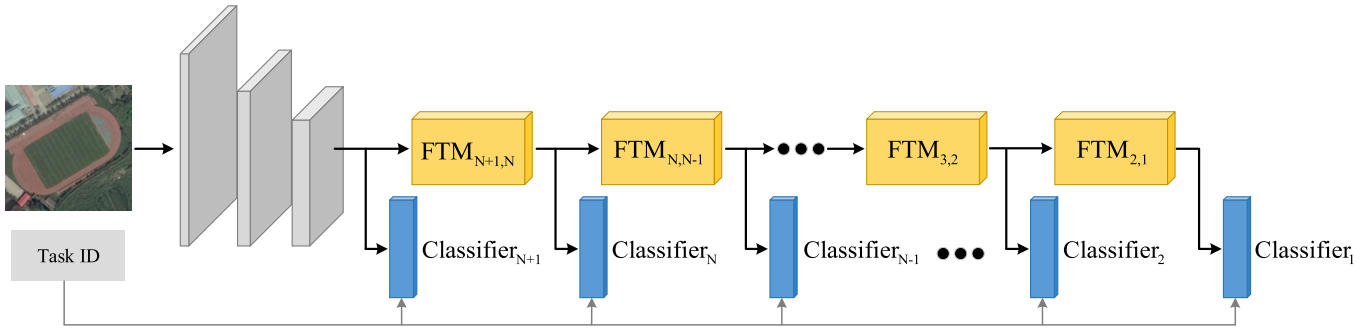


Fig. 4. Model of LIL after learning $N+1$ tasks.

to imitate the features extracted by the TCE of the memory model in the feature space of the old task at the semantic level.

After the training for the new task, the memory model and the discriminator are discarded, and only the new model is retained. The TCE of the new model learns to extract features in the feature space of the new task. The module $\text{Classifier}_{\text{new}}$ has the ability to classify the features of the new task. The FTM is able to transfer features from the new task to the old task. Thus, the new model is qualified for classifying remote sensing images of both the old and new tasks.

The prediction process is given as follows. First, input one remote sensing image and its task ID (new or old task). Second, the features are extracted by the TCE in the feature space of the new task. If the input image belongs to the new task, $\text{Classifier}_{\text{new}}$ will be used to classify the features directly. If it is an image of the old task, the extracted features will be transferred from the new task to the old task through the FTM, and $\text{Classifier}_{\text{old}}$ will be used to achieve classification according to the transferred features.

When the network continues to learn new tasks, repeat the above steps to realize incremental learning. In brief, the existing model is copied and fixed as a memory network. Then, an FTM and a classifier are expanded on the basis of the existing model to obtain a new model. Finally, the classification loss and the dual-constraint loss are applied to train the new model on the data of new tasks. When the model that has learned N tasks needs to learn *Task* $N+1$, the specific process of learning is shown in Algorithm 1.

The architecture of the model in LIL after learning $N+1$ tasks is shown in Fig. 4. The model contains one TCE and N FTMs $\text{FTM}_{m,m-1}$, where $2 \leq m \leq N+1$ and $N+1$ are task-specific classification modules Classifier_n, where $1 \leq n \leq N+1$. The mission of $\text{FTM}_{m,m-1}$ is to map the features from *Task* m to *Task* $m-1$ in the feature spaces. Classifier_n classifies the corresponding features of *Task* n to obtain the predictions of input images. The procedure of using the model to predict the input remote sensing images is given as follows. The network receives an image and its task ID. Then, the TCE extracts features in the feature space of *Task* $N+1$. The classification module Classifier_{N+1} will be used immediately to output results if the task ID is $N+1$. Otherwise, LIL sends the extracted features to FTMs successively, until the features are mapped to the feature space of the task that the input image belongs to. Finally, the task-specific classification module $\text{Classifier}_{\text{taskID}}$ is used to classify the converted features.

Algorithm 1 Process of Learning a New Task (*Task* $N+1$) in LIL

Input: The input images $\boldsymbol{x}$ and corresponding labels $\boldsymbol{y}$ of *Task* $N+1$, the weights $\boldsymbol{\theta}_{\text{old}}$ of the old model that performs well on previous N tasks, hyper parameters λ_1, λ_2, learning rates $\boldsymbol{l}_m$, $\boldsymbol{l}_d$ for the model and the discriminator, and the number of epochs $\boldsymbol{n}$.

Output: The weights $\boldsymbol{\theta}_{\text{new}}$ of the new model that can classify both new and old tasks.

Establish a memory model based on $\boldsymbol{\theta}_{\text{old}}$.
Extend $\text{FTM}_{N+1,N}$ and Classifier_{N+1} on the old model to generate a new model $\boldsymbol{\theta}_{\text{new}}$.
Build a discriminator and initialize its parameters $\boldsymbol{\theta}_d$.
for $\boldsymbol{i} = \mathbf{1}$ to $\boldsymbol{n}$ **do**
    Calculate $\mathcal{L}_{\text{task}}$ using the outputs of the new model.
    Calculate $\mathcal{L}_{\text{dist}}$ using the outputs of both the new model and the memory model.
    Calculate $\mathcal{L}_{\text{adv}}$ for $\boldsymbol{\theta}_{\text{new}}$ using the outputs of the discriminator.
    $\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda_1 \mathcal{L}_{\text{dist}} + \lambda_2 \mathcal{L}_{\text{adv}}$.
    Update $\boldsymbol{\theta}_{\text{new}} \leftarrow \boldsymbol{\theta}_{\text{new}} - \boldsymbol{l}_m \frac{\partial \mathcal{L}}{\partial \boldsymbol{\theta}_{\text{new}}}$.
    Calculate $\mathcal{L}_{\text{adv}}$ for the discriminator $\boldsymbol{\theta}_d$.
    Update $\boldsymbol{\theta}_d \leftarrow \boldsymbol{\theta}_d - \boldsymbol{l}_d \frac{\partial \mathcal{L}_{\text{adv}}}{\partial \boldsymbol{\theta}_d}$.
end for
Discard old model, memory model, and discriminator.

In general, LIL realizes incremental learning with a small number of parameters and no need for extra memory for preserving the data of previous tasks. It is simple yet surprisingly forceful in mitigating catastrophic forgetting with slightly increased parameters as the number of learned tasks increases.

B. Task-Common Feature Extractor

In order to solve the problem that the backbones of existing parameter expansion-based incremental learning models are heavy, the mission of feature extractors in conventional methods is modified in our proposed LIL. The feature extractor in LIL only extracts high-level semantic features from images in the feature space of the latest learned task instead of the

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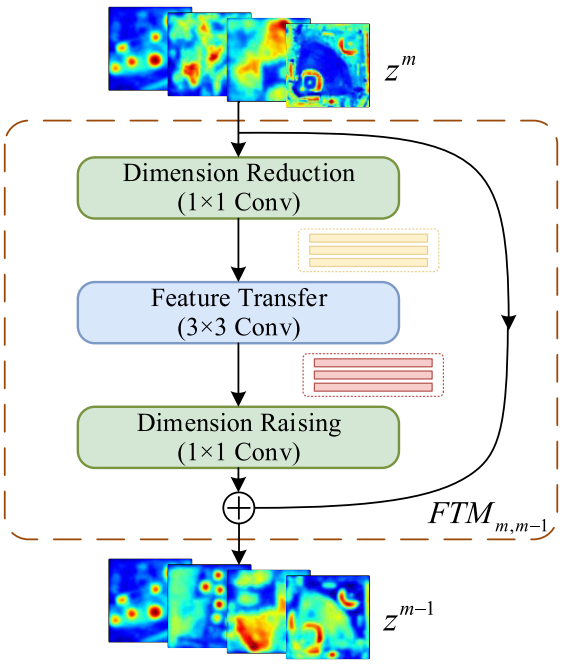


Fig. 5. Structure of the FTM $\mathrm{FTM}_{m,m-1}$.

task-sharing features among all tasks. Besides, the features extracted by it are also used in common for all tasks. Hence, the feature extractor in LIL is termed TCE. The formula of TCE is given as follows:

$$z = f(\boldsymbol{x}) \tag{1}$$

where $\boldsymbol{x}$ is the input image with size $C_{\text{in}} \times H \times W$, and z is the extracted feature map whose size is $C_{\text{out}} \times (H/r) \times (W/r)$. Note that C_{in} and C_{out} represent the number of channels of the input image and the output feature map, respectively. H and W are height and width of the image. r is the downsampling stride of the TCE.

Different from existing incremental learning methods, with no need for extracting features shared among all tasks, the TCE only extracts features in the feature space of the last learned task. Therefore, it is able to apply a backbone with fewer parameters than the TCE to reduce parameters in LIL.

C. Feature Transfer Module

For the purpose to lighten the structures of task-specific modules in parameter expansion-based methods, a novel FTM is proposed to transform features between two adjacent tasks. Through FTMs, the features output by the TCE in the feature space of the last task are able to be used for all previous tasks.

Distinct from the task-specific modules in existing approaches, FTMs in LIL perform on the domain of features instead of images. They learn the mapping functions of data distributions between different tasks in the feature space to realize the transferring of features from new tasks to adjacent old tasks.

The input and the output of an FTM are feature maps with the same size, but the feature spaces in which they are located belong to different tasks. The function of $\mathrm{FTM}_{m,m-1}$ is to map features from the feature space of *Task* m to the feature space of *Task* $m-1$, which is described as follows:

$$z^{m-1} = \mathrm{FTM}_{m,m-1}\left(z^m\right) \tag{2}$$

where z^{m-1} and z^m separately represent the feature maps in the feature spaces of *Task* $m-1$ and *Task* m.

The FTM only processes extracted high-level features, which can be designed to be lightweight. Only a small quantity of convolutional layers is applied to implement the transformation between features. In addition, we draw on the idea of residual network [85] to make the FTM only learn the differences of features between two adjacent tasks, which enables us to learn the mapping relationship effectively.

The structure of the FTM $\mathrm{FTM}_{m,m-1}$ is shown in Fig. 5. One 1×1 convolutional layer is used for dimensionality reduction to further reduce the parameters of the module. Then, three 3×3 convolutional layers are applied to learn the function for feature transfer. Afterward, another 1×1 convolutional layer is added to increase the dimension of the feature map to the same as the input. Finally, add the input feature map z^m to the obtained result for outputting the final transferred feature z^{m-1}.

In some cases, it is necessary to execute continuous transfer of features, such as transferring from the newly learned *Task* $N+1$ to the first learned *Task* 1. Since the input and the output feature maps of FTMs have the same size, the adjacent FTMs are qualified to be connected in series to transfer features from the feature space of one task to any previous tasks, as shown in Fig. 4. The formula for sequential mapping of FTMs is presented as follows:

$$z^m = \mathrm{FTM}_{m+1,m}, \ldots, \mathrm{FTM}_{n-1,n-2}\mathrm{FTM}_{n,n-1}\left(z^n\right) \tag{3}$$

where $m < n$.

In addition, each FTM is followed by a task-specific classification module $\mathrm{Classifier}_n$, as shown in Fig. 4. It predicts the final classification results according to the features in the feature space of the corresponding task (*Task* n). As illustrated in (4), the input of $\mathrm{Classifier}_n$ is a feature map in the feature space of *Task* n with the size of $C_{\text{out}} \times (H/r) \times (W/r)$, and the output $\boldsymbol{y}^n$ is the predicted probabilities of categories in *Task* n

$$\boldsymbol{y}^n = \mathrm{Classifier}_n\left(z^n\right). \tag{4}$$

When learning a new task, only an FTM and a classification module need to be expanded to the existing model. Since both modules operate on high-level features, they are able to be designed to be compact. Thus, as the number of tasks increases, only a handful of parameters are expanded in LIL to achieve incremental learning.

D. Dual-Constraint Loss

Existing incremental learning models usually use distillation loss for distilling the knowledge of an old model into a new model to overcome catastrophic forgetting. When distillation loss is used alone to train FTMs of LIL, we find that the forgetting of old knowledge is indeed reduced, but the mACC of the model among all tasks is still far behind JT, which

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is considered as the upper bound of the performance of incremental learning models.

Distillation loss only constrains features in terms of numerical values, which we call feature-level restriction. It expects that the features transferred by the newly added FTM of the new model are exactly the same as the features extracted by the old model. As we all know, the features extracted by the backbone contain high-level semantic information. Only constraining features at the feature level is not enough to equip FTMs to learn the mapping relationships between features of different tasks. Therefore, in order to make full use of the semantic information of extracted features, inspired by GAN [45], dual-constraint loss with knowledge distillation and adversarial learning is put forward to promote the learning of FTMs.

In LIL, dual-constraint loss supervises FTMs to map and align features at both feature level and semantic level. On the one hand, distillation loss $\mathcal{L}_{\text{dist}}$ is applied to train FTMs at the feature level. On the other hand, through adversarial training, a discriminator and adversarial loss $\mathcal{L}_{\text{adv}}$ are proposed to facilitate the learning of FTMs at the semantic level.

1) Distillation Loss: In order to enable FTMs to convert features effectively, distillation loss $\mathcal{L}_{\text{dist}}$ is used to constrain the modules at the feature level. As shown in Fig. 3, the inputs of $\mathcal{L}_{\text{dist}}$ are the feature maps extracted by the TCE of the memory model and the feature maps transferred by the newly added FTM of the new model for the same image. We use distillation loss to calculate the numerical differences between the two feature maps at the feature level.

The goal of training FTMs is to minimize $\mathcal{L}_{\text{dist}}$ to reduce the numerical discrepancies between two types of features. In the subsequent ablation experiments in this article (see Section IV-E3), distillation loss is experimented with mean-square error (MSE) loss [see (5)] and cosine similarity (CS) loss [see (6)]

$$\mathcal{L}_{\text{dist-MSE}} = \mathbb{E}_{\left(z_k^n, \hat{z}_k^n\right)} \left(z_k^n - \hat{z}_k^n\right)^2 \tag{5}$$

$$\begin{aligned} \mathcal{L}_{\text{dist-CS}} &= \mathbb{E}_{\left(z_k^n, \hat{z}_k^n\right)} \left(1 - \cos\left\langle z_k^n, \hat{z}_k^n \right\rangle\right) \\ &= \mathbb{E}_{\left(z_k^n, \hat{z}_k^n\right)} \left(1 - \frac{z_k^n \cdot \hat{z}_k^n}{\left\|z_k^n\right\| \left\|\hat{z}_k^n\right\|}\right) \end{aligned} \tag{6}$$

where $\mathbb{E}$ means the average value of all input images. z_k^n and $\hat{z}_k^n$ are the feature maps extracted by the memory model and transferred by the FTM of the new model according to the kth image in *Task* n, respectively. $\langle z_k^n, \hat{z}_k^n \rangle$ in (6) represents the angle between two feature maps.

The experimental results prove that CS loss is more effective than MSE loss in guiding the learning of FTMs. Therefore, unless otherwise specified, $\mathcal{L}_{\text{dist-CS}}$ is used as the distillation loss in consequent experiments.

2) Adversarial Loss: Distillation loss only constrains FTMs at the feature level. In other words, it only pays attention to numerical information without semantic information of features. Illuminated by GAN, a discriminator is designed in LIL when training the new model. It forms an adversarial training framework with the newly added FTM and the TCE.

In the adversarial training framework, expect the discriminator, the TCE and the FTM are served as the generator.

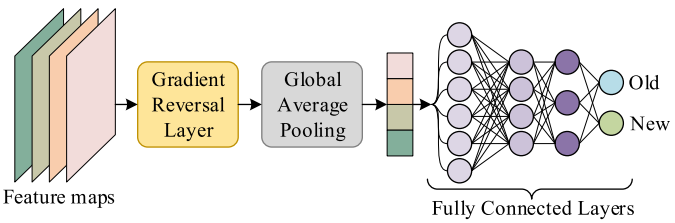


Fig. 6. Structure of the discriminator for adversarial loss in dual-constraint loss.

The essence of the discriminator is a binary-classifier to differentiate features at the semantic level. It distinguishes whether the input feature map is extracted by the TCE of the memory model or extracted and transferred by the newly expanded FTM of the new model. The adversarial loss $\mathcal{L}_{\text{adv}}$ is introduced for the training of the generator and the discriminator. As shown in (7), the cross-entropy loss is used for adversarial loss

$$\mathcal{L}_{\text{adv}} = \max_{G} \min_{D} - \mathbb{E}_{(z_k, t_k)} t_k \log(D(z_k)) \tag{7}$$

where G and D represent the generator (including the TCE and the newly added FTM) and the discriminator, respectively. z_k is the feature map of the kth image. t_k represents the corresponding model label of the input feature map, that is, whether the feature belongs to the memory model or the new model.

The discriminator minimizes $\mathcal{L}_{\text{adv}}$ to distinguish the two kinds of features as much as possible, while the TCE and the newly added FTM maximize $\mathcal{L}_{\text{adv}}$ to deceive the discriminator so that the discriminator cannot accurately confirm the origin of features. Through adversarial training, the newly added FTM learns to imitate the features extracted by the memory model in the feature space of the previous learned task at the semantic level as much as possible.

The structure of the discriminator is shown in Fig. 6. Same as the distillation loss, the inputs of the discriminator are also the feature maps output by the TCE of the memory model and transferred by the FTM of the new model. In order to achieve end-to-end adversarial training, the gradient reversal layer (GRL) [86] is used first in the discriminator. On the one hand, during forward propagation, GRL implements no operation, and its output is equal to the input. On the other hand, the gradients of parameters are reversed during backward propagation to achieve the purpose of maximizing the adversarial loss for the generator ($\mathcal{L}_{\text{adv}-G} = -\mathcal{L}_{\text{adv}}$). Then, the feature maps are pooled into feature vectors through the global average pooling (GAP) layer. Finally, several fully connected (FC) layers are added to output binary classification results based on the feature vectors.

When the model learns a new task, in addition to dual-constraint loss, the cross-entropy loss shown in (8) is used as classification loss $\mathcal{L}_{\text{task}}$ to train the classification module for the new task

$$\mathcal{L}_{\text{task}} = -\mathbb{E}_{\left(x_k^n, y_k^n\right)} \sum_{c=1}^{C} y_{k,c}^n \log \hat{y}_{k,c}^n \tag{8}$$

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where $(\boldsymbol{x}_k^n, \boldsymbol{y}_k^n)$ represent the kth input image belonging to *Task* n and its ground-truth label. C is the number of categories in *Task* n. $y_{k,c}^n$ and $\hat{y}_{k,c}^n$ are true probabilities and predicted probabilities of the kth input image of *Task* n on the cth category, respectively.

Combining classification loss and dual-constraint loss, the total loss in LIL is given as follows:

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda_1 \mathcal{L}_{\text{dist}} + \lambda_2 \mathcal{L}_{\text{adv}} \tag{9}$$

where λ_1 and λ_2 are balance factors used to control the impact of each component. In the following experiments, unless otherwise stated, the default values of λ_1 and λ_2 are set to 1.

IV. Experiments

In this section, we implement sufficient experiments to evaluate the performance of LIL. First, we briefly introduce the datasets for remote sensing image scene classification used in experiments. Then, the evaluation metrics and implementation details of LIL and other compared methods are explained. Finally, the incremental learning performance of LIL and compared methods on a single dataset and a sequence of multiple datasets is analyzed, which proves that LIL with lightweight architecture is superior to state-of-the-art incremental learning approaches in overcoming catastrophic forgetting.

A. Datasets

AID [52] is used for main experiments in this article. In addition, we also conduct experiments on two large-scale datasets (EuroSAT [57] and BigEarthNet [58]) and a sequence of seven datasets (AID, EuroSAT, BigEarthNet, RS_C11 [53], RSSCN7 [54], SIRI-WHU [55], and SAT-4 [56]) to verify the performance of LIL in complex scenarios for incremental learning.

1) AID: The images in AID are collected on Google Earth and mainly contain 30 categories. The number of images for each category ranges from 220 to 420. AID contains a total of 10 000 images with the size of 600 × 600 pixels. Since the images on Google Earth are collected by different remote sensing sensors, the data in AID are multisource and multiresolution. The pixel resolution changes from about 8 m to about half a meter. Moreover, the images in AID cover many areas in different countries and are captured at different seasons under various imaging conditions, which increases the differences of distributions between images. Thus, AID is suitable for the studies of incremental learning. In order to make the dataset applicable for incremental learning settings, we divide it into ten parts, and each part represents one task containing three categories, called ten-split AID.

2) EuroSAT: EuroSAT is a dataset for the task of land use and land cover classification using Sentinel-2 satellite images. There are total 27 000 labeled and georeferenced images distributed in 34 European countries in the dataset. The image patches in EuroSAT are multispectral samples covering 13 spectral bands and measure 64 × 64 pixels. There are ten different land use, and land cover classes are annotated in the dataset. EuroSAT is available in RGB and multispectral versions. In the experiment, we use the RGB version and divide it into two parts evenly according to the number of categories, called two-split EuroSAT.

3) BigEarthNet: BigEarthNet is a large-scale multilabel dataset. It consists of 590 326 Sentinel-2 images with 12 different spectral bands (the tenth band without surface information is excluded). The size of each band is one of 120 × 120 pixels, 60 × 60 pixels, and 20 × 20 pixels. The data in BigEarthNet are acquired from 125 Sentinel-2 tiles from ten different European countries. The dataset contains a total of 43 categories provided from the CORINE Land Cover, and the number of categories associated with each image varies between 1 and 12. We preprocessed the data for the incremental learning experiment, and the specific operation details are described in Section IV-G.

4) RS_C11: RS_C11 is collected from several cities in the United States on Google Earth, such as Los Angeles, New York, Washington DC, San Francisco, San Diego, Chicago, and Houston. It contains 11 categories. The number of images for each category is about 100. There are 1232 images with the resolution of 0.2 m in RS_C11. Each image has the size of 512 × 512 pixels.

5) RSSCN7: RSSCN7 includes 2800 remote sensing images collected from Google Earth. The images are labeled into seven categories. There are 400 images with four different scales for each category. Each image has the size of 400 × 400 pixels.

6) SIRI-WHU: SIRI-WHU mainly contains 2400 remote sensing images of urban areas in China. The images are collected on Google Earth. There are 12 categories in SIRI-WHU. Each category contains 200 images with 2-m spatial resolution. Each image measures 200 × 200 pixels.

7) SAT-4: The images in SAT-4 are sampled from the National Agriculture Imagery Program (NAIP) dataset. SAT-4 consists of a total of 500 000 images involving four bands of red, green, blue, and near-infrared (NIR). The images are divided into four broad land cover classes, and the size of each image is 28 × 28 pixels.

B. Evaluation Metrics

Same as [44], the model is evaluated on all previously learned tasks after training each new task. We use mACC [6] and BackWard Transfer (BWT) [44] as evaluation metrics to appraise the performance of LIL and other compared incremental learning approaches. In addition, we also calculate the number of parameters in models to evaluate the memory spaces required by models.

mACC represents the mACC of a model across all tasks after training the last task. It examines the average performance of an incremental learning model on total tasks and reflects the balance effects of the model between learning new tasks and retaining the knowledge of previous tasks. The calculation formula of mACC is shown in (10). The larger the mACC, the better the performance of the incremental learning model

$$\text{mACC} = \frac{1}{T} \sum_{i=1}^{T} P_{T,i} \tag{10}$$

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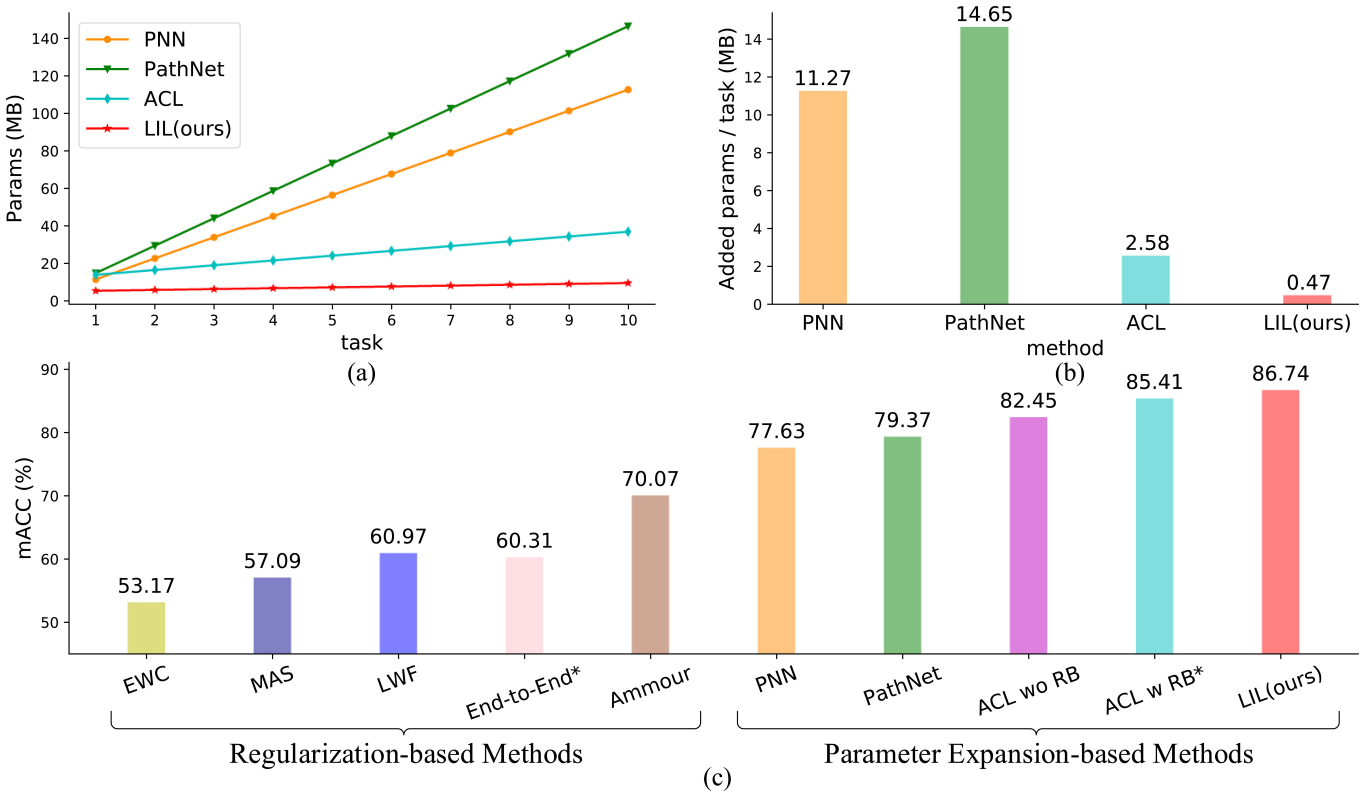


Fig. 7. Results of state-of-the-art incremental learning methods and our proposed LIL on the ten-split AID dataset. (a) Number of parameters of parameter expansion-based models after learning each task. (b) Average number of parameters that need to be added when learning a new task. (c) mACC of incremental learning methods on all tasks. * indicates that the method applies data replay.

where T is the total number of tasks for incremental learning and $P_{j,i}$ represents the classification accuracy on *Task* i after learning *Task* j.

BWT is used to measure the forgetting of a model on old tasks after learning new tasks. BWT is calculated as follows:

$$\text{BWT} = \frac{1}{T-1}\sum_{i=1}^{T-1}\left(P_{i,i} - P_{T,i}\right). \tag{11}$$

When BWT $>$ 0, it indicates that the performance of old tasks is affected when the model learns new tasks, that is, the model faces catastrophic forgetting. Conversely, when BWT $<$ 0, it means that the effects of old tasks are promoted when the model is trained on new tasks. Thus, for incremental learning models, the smaller the BWT, the less the model has forgotten on previous learned tasks.

C. Implementation Details

As mentioned in Section III-B, the TCE in LIL only extracts features related to the last learned task. It can be designed to be compact. Thus, we use the first four stages of Resnet18 [85] (i.e., Conv 1–4) as the TCE. In addition, the 7 $\times$ 7 convolutional layer in Conv1 is replaced with three 3 $\times$ 3 convolutional layers with stride 1. The max-pooling layer behind Conv1 is canceled. Therefore, the downsampling stride of the TCE in LIL is 4, and the output feature map has 256 channels. The FTM is composed of two 1 $\times$ 1 convolutional layers and three 3 $\times$ 3 convolutional layers. The task-specific classification module is composed of a GAP layer followed by three FC layers. As shown in Fig. 6, the discriminator consists of a GRL, a GAP layer, and three FC layers.

To demonstrate the advantages of our method, LIL is compared with some basic methods (FT and JT) and state-of-the-art incremental learning approaches, including regularization-based methods (EWC [20], MAS [22], LWF [16], End-to-End [18], and Ammour *et al.* [29]) and parameter expansion-based methods (PNN [42], PathNet [43], and ACL [44]). For a fairer comparison, the full Resnet18 with the replaced Conv1 is used as the backbone for these methods, and task IDs of images are also available during training and testing.

As the input of the network, the images in AID, BigEarthNet, RS_C11, RSSCN7, and SIRI-WHU are all resized to 200 $\times$ 200 pixels, and the images in EuroSAT and SAT-4 are resized to 64 $\times$ 64 pixels. During training, we set batchsize to 16 and use stochastic gradient descent (SGD) optimizer to train 100 epochs for each task. The initial learning rate is set to 0.001 with the decay of ten times in the 40th, 70th, and 90th epochs. Momentum and weight decay coefficients are 0.9 and 0.0001, respectively. No data augmentation is used during training and testing, and all experimental results are obtained without bells and whistles. Each experiment is run three times. For each time, the AID dataset is randomly and evenly divided according to categories as a form of ten-split AID to generate different task sequences for incremental learning. The mean

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TABLE I
INCREMENTAL LEARNING RESULTS ON TEN-SPLIT AID

Method	$mACC(\%)$	$BWT(\%)$
JT	82.75	-
FT	50.97±5.34	46.18±5.51
EWC [20]	53.17±6.92	43.55±8.34
MAS [22]	57.09±7.07	34.35±10.09
LWF [16]	60.97±20.16	26.45±16.62
End-to-End* [18]	60.31±1.22	24.36±5.19
Ammour *et al.* [29]	70.07±1.32	24.51±1.33
PNN [42]	77.63±4.79	-
PathNet [43]	79.37±5.27	4.38±3.13
ACL wo RB [44]	82.45±4.65	5.55±5.58
ACL w RB* [44]	85.41±1.52	3.19±2.09
LIL (ours)	**86.74±2.75**	**3.03±2.16**

TABLE II
RESULTS OF ABLATION STUDY FOR TCE

Method	Backbone	$mACC(\%)$	$BWT(\%)$
ACL w RB*	part_resnet18	62.83±9.88	27.42±6.44
	resnet18	85.41±1.52	3.19±2.09
LIL (ours)	part_resnet18	86.74±2.75	**3.03±2.16**
	resnet18	**86.95±3.33**	3.07±2.09

TABLE III
RESULTS OF ABLATION STUDY FOR DUAL-CONSTRAINT LOSS

#	$\mathcal{L}_{task}$	$\mathcal{L}_{dist}$ MSE	$\mathcal{L}_{dist}$ CS	$\mathcal{L}_{adv}$	$mACC(\%)$	$BWT(\%)$
1	✓				47.50±10.89	45.75±5.55
2	✓	✓			80.21±3.36	4.32±4.17
3	✓		✓		82.55±1.47	3.88±1.50
4	✓			✓	63.37±5.63	35.26±6.04
5	✓	✓		✓	82.76±3.33	4.11±5.55
6	✓		✓	✓	**86.74±2.75**	**3.03±2.16**

and standard deviation of mACC and BWT are presented as final results. All experiments are performed on one RTX-2080Ti GPU.

D. Experiments on Ten-Split AID

In the experiments on ten-split AID, we measure mACC, BWT, and the number of parameters of our proposed LIL and advanced incremental learning approaches. As shown in Table I and Fig. 7, LIL with the least parameters achieves state-of-the-art performance among all reported incremental learning methods.

In Table I, the first row represents two basic methods (with no incremental learning behavior), including JT and FT, the second row shows regularization-based methods, and the third row contains parameter expansion-based methods. Among them, * indicates that the method calls data replay with saving five images for each category of old tasks.

FT means that the model is only trained on the data of new tasks without carrying out any behavior for retaining old knowledge. FT achieves BWT $= 46.18\% \pm 5.51\%$, indicating that severe catastrophic forgetting has occurred. Besides, JT does not meet the settings of incremental learning. It implies that the data of all tasks are learned at the same time, and there is no forgetting for JT. Under normal circumstances, JT is considered as the upper bound of incremental learning methods. However, compared with JT, ACL w RB (ACL with data replay) and LIL are 2.66% and 3.99% more in mACC, respectively. We consider that, as one of the inputs in these incremental learning methods, task IDs provide the prior knowledge of the data, which simplifies the difficulties of tasks and improves performance.

Among the incremental learning methods based on regularization, the method that Ammour *et al.* [29] proposed shows better performance than others. Due to the lack of scalability and insufficient constraints, the forgetting rates of these methods are still high. Besides, as shown in Table I and Fig. 7(c), it is observed that the performance of parameter expansion-based methods is generally superior to that of regularization-based methods. It proves that parameter expansion-based methods increase parameters to improve the learning ability of models, which is beneficial to incremental learning.

For parameter expansion-based approaches, PNN generates a complete single-column model for each task, and there is no forgetting, but it is difficult to train the model due to the complex architecture, so the mACC of PNN is slightly low. PathNet softly improves the performance through path selection, but its model is even larger than PNN. ACL improves the precision of tasks while reducing the forgetting rate. However, compared with ACL wo RB (ACL without data replay), LIL has a 4.29% higher in mACC and a 2.52% lower in BWT in the absence of data replay. Furthermore, ACL w RB (ACL with data replay) has the preferable fulfillment for incremental learning than ACL wo RB. Among all compared methods, ACL w RB achieves excellent performance, where mACC $= 85.41\% \pm 1.52\%$ and BWT $= 3.19\% \pm 2.09\%$. Nevertheless, our proposed LIL without storing the data of old tasks reaches the higher score of mACC $= 86.74\% \pm 2.75\%$ and BWT $= 3.03\% \pm 2.16\%$ than ACL w RB. These results prove the effectiveness of our proposed LIL for incremental learning in the field of remote sensing image scene classification.

In terms of the number of parameters, due to the changes in the role of the TCE in LIL, we remove the last stage of Resnet18. Therefore, compared with JT, FT, and regularization-based methods that use the whole Resnet18 as backbones, the number of parameters in LIL is reduced by more than 7 MB. As mentioned earlier, the task-specific modules in existing parameter expansion-based models are

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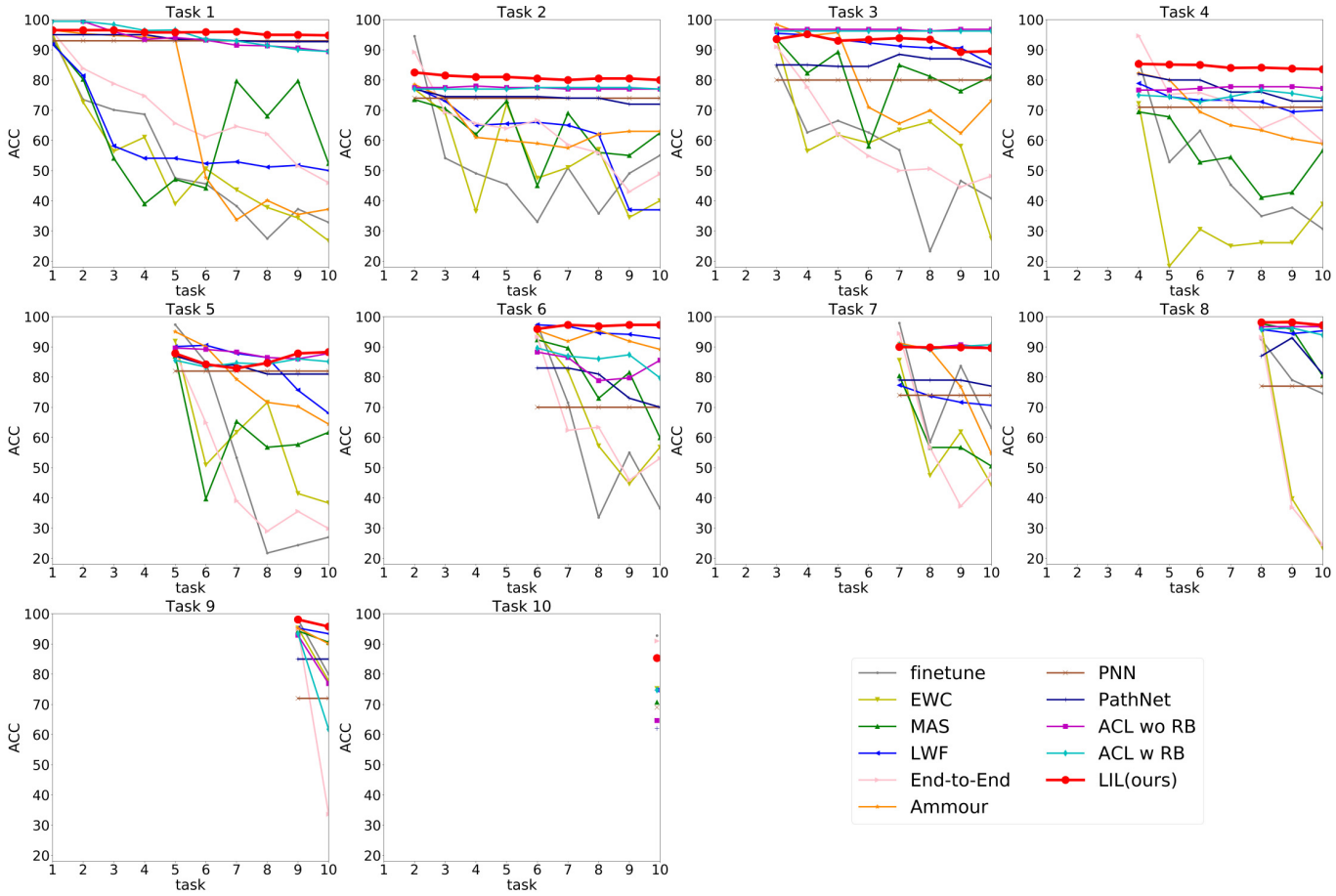


Fig. 8. Accuracy of incremental learning approaches after learning each task of ten-split AID. Each subgraph shows the changing trend of the accuracy for a specific task when the model is trained on the data of subsequent tasks successively.

applied in the domain of images, which causes the parameters in models to rise sharply as the amount of tasks increases. In contrast, because FTMs process on the domain of features, the proposed LIL only needs to add a few parameters to the model when a new task comes. As presented in Fig. 7(a) and (b), PNN and PathNet have a huge amount of parameters because they expand an independent branch for each task. After learning ten tasks, the architectures of PNN and PathNet both contain more than 100-MB parameters. The advanced method, ACL, contains 36.95-MB parameters, while LIL only contains 9.55-MB parameters for its model. Hence, the performance among all tasks of LIL exceeds the advanced method ACL under the condition that the number of parameters is reduced by about 74% of ACLs. These comparison results illustrate that LIL enables the model to adapt to scenario changes automatically by adding a handful of parameters for new tasks.

Fig. 8 shows the results of the accuracy on LIL and compared methods for all previous learned tasks when each task is trained. As shown in Fig. 8, LIL has the distinguished performance on most tasks of ten-split AID and reduces catastrophic forgetting to the greatest extent. Furthermore, it is found that, when learning *Tasks* 8, 9, and 10, the accuracy of *Task* 5 in LIL is promoted. It indicates that, when learning subsequent tasks, the TCE and the FTMs may extract and map some knowledge that has been forgotten after learning some tasks or has not been learned before, which facilitates the performance of previous tasks.

Besides, it can be seen from Fig. 8 that, similar to fine-tuning, the accuracy of EWC and MAS fluctuates greatly. Meanwhile, LWF, End-to-End, and Ammour also forget a good deal of knowledge of old tasks when learning new tasks. It intuitively shows that the methods based on regularization are less constrained and have great impacts on previous tasks when learning new tasks. As an advanced incremental learning method based on parameter expansion, ACL achieves high accuracy and a low forgetting rate. However, our proposed LIL achieves more outstanding accuracy and forgetting rate than ACL under the condition that the amount of parameters is about one-fourth of the parameters in ACL.

In summary, experimental results on ten-split AID show that LIL realizes an LIL model in the field of remote sensing image scene classification. Under the situation of occupying the least memory, LIL without the need for storing the data of old tasks achieves state-of-the-art performance for incremental learning.

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TABLE IV
RESULTS OF ABLATION STUDY FOR BALANCE FACTORS IN LOSS

Balance Factor		λ_1					
		0.1		1		5	
		$mACC(\%)$	$BWT(\%)$	$mACC(\%)$	$BWT(\%)$	$mACC(\%)$	$BWT(\%)$
λ_2	0.1	79.53±3.12	12.53±2.16	84.57±2.98	3.79±2.74	85.70±2.39	3.68±2.45
	1	82.49±2.53	8.26±2.24	**86.74±2.75**	3.03±2.16	83.65±2.36	2.31±1.72
	5	83.98±2.94	4.35±1.80	86.01±3.03	2.93±2.12	84.04±3.17	**1.35±1.43**

TABLE V
RESULTS OF ABLATION STUDY FOR THE NUMBER OF TASKS

Number of Tasks	3		5		10		15	
Method	$mACC(\%)$	$BWT(\%)$	$mACC(\%)$	$BWT(\%)$	$mACC(\%)$	$BWT(\%)$	$mACC(\%)$	$BWT(\%)$
ACL w RB*	58.20±2.33	**0.03±0.23**	70.34±2.12	1.85±1.64	85.41±1.52	3.19±2.09	84.73±2.81	4.33±1.66
LIL (ours)	**86.68±0.85**	0.05±0.55	**86.56±1.77**	**1.74±1.26**	**86.74±2.75**	**3.03±2.16**	**85.58±5.39**	**4.02±1.06**

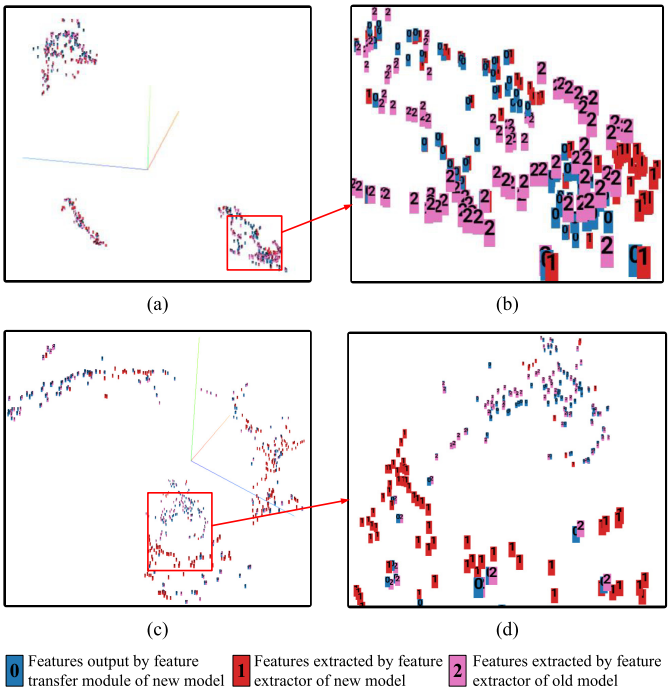


Fig. 9. Visualizing results of features obtained from the data of old task. (a) and (b) When the new task is similar to the old task. (c) and (d) When the new task is different from the old task.

E. Ablation Studies

In order to demonstrate the effectiveness of the proposed LIL and its components, we conduct sufficient ablation experiments on the AID dataset. In addition, we verify the performance of LIL in a more complex incremental learning environment, that is, performing experiments on five different datasets, each of which represents one task.

1) Ablation Study for Task-Common Feature Extractor: To verify the effectiveness of the TCE of LIL in simplifying the backbone, ablation experiments on ten-split AID are implemented. According to the results in Table I, we choose ACL w RB, achieving the impressive performance on ten-split AID, as the compared method.

As for LIL and ACL w RB, we use the whole Resnet18 except for the last FC layer and part of Resnet18 (Conv $1-4$) as feature extractors to perform experiments on the ten-split AID dataset. Table II shows the experimental results. The results serve to show that, when the feature extractor of ACL w RB is shrunk from the complete Resnet18 to a part of Resnet18, the performance of the model is radically reduced, where mACC drops by 22.58% and BWT increases by 24.23%. It indicates that, as task-sharing modules, the feature extractors of traditional parameter expansion-based incremental learning methods are incapable to effectively extract task-sharing features from images of all tasks when their structures become compact. Therefore, the large-capacity networks must be applied as the task-sharing feature extractors.

On the contrary, when the two kinds of backbones are used as the TCE in LIL, the performance of the model hardly changes. Since the TCE in LIL only extracts features in the feature space of a single task, a light network is sufficient to complete the mission of feature extraction. Therefore, the TCE can be implemented with a compact structure. It verifies that the TCE proposed in LIL can effectively lighten the structure of the backbone while maintaining the preeminent performance on feature extraction.

2) Ablation Study for Feature Transfer Module: Here, we describe the role of the FTM in LIL by using the t-distributed Stochastic Neighbor Embedding (t-SNE) [87] plots for experiments on ten-split AID. Fig. 9 visualizes three kinds of features for the data of an old task, including the features transferred by the FTM of the new model, the features extracted by the TCE of the new model (i.e., without feature transfer), and the features output by the TCE of the old model.

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In Fig. 9(a) and (b), we show the visualizations of features when the data of the new task are similar to that of the old task. After the model learns the new task, the features extracted by the TCE of the new model from the data of the old task can still be clearly divided into three clusters (that is, three categories of the old task). The model does not produce catastrophic forgetting. Moreover, it can be seen from the partial enlarged view [see Fig. 9(b)] that the features transformed by the FTM are closer to the features extracted by the old model than the features extracted by the TCE of the new model. It indicates that the FTM is able to map features from the feature space of the new task back to the feature space of the old task.

As shown in Fig. 9(c) and (d), there are no obvious dividing lines for the features output by the TCE of the new model when the data of the new task differ greatly from the data of the old task. Thus, the classifier of the old task is unable to classify the features accurately, resulting in catastrophic forgetting. However, through the FTM, the features extracted in the feature space of the new task are remapped back to the feature space of the old task, which alleviates the issue of forgetting.

By visualizing the features, we prove the effectiveness of the proposed light FTM. It is qualified to transfer features from the feature space of a new task back to the feature space of the adjacent old task even if the differences of data between two tasks are violent.

3) Ablation Study for Dual-Constraint Loss: In order to verify the effect of dual-constraint loss proposed in LIL for training FTMs, we conduct multiple experiments on the ten-split AID dataset, as shown in Table III.

Ablation experiment #1 indicates that there is only classification loss $\mathcal{L}_{\text{task}}$ used to train the model, and FTMs are merely and randomly initialized without training. The performance of the model in #1 is lower than the performance of FT in Table I, which may be caused by the random initialization of FTMs. By comparing the ablations #1, #2, and #3, it is found that the distillation loss $\mathcal{L}_{\text{dist}}$ at the feature level effectively supervises the training of FTMs; both mACC and BWT are greatly improved. When distillation loss is used alone, the performance of CS loss, achieving mACC $= 82.55 \pm 1.47\%$ and BWT $= 3.88 \pm 1.50\%$, is better than MSE loss.

Observing ablation experiments #1 and #4, it can be seen that adversarial loss $\mathcal{L}_{\text{adv}}$ also plays an import role in training FTMs. Since the binding force of adversarial loss is not as strong as the distillation loss, which requires the same value of features, the effect of using adversarial loss solely is slightly worse than only applying distillation loss. Through ablation experiments #2 and #5 (or #3 and #6), it can be seen that applying dual-constraint loss with both distillation loss and adversarial loss is better than using one loss function alone. By comparing the ablations #5 and #6, the same as when using distillation loss only, the performance of using CS loss for distillation loss is still better than that of MSE loss in the case of applying dual-constraint loss. In consequence, the dual-constraint loss with distillation loss of CS loss and the adversarial loss of cross-entropy loss achieves the excellent performance, which is used in the remaining experiments.

4) Ablation Study for Balance Factors in Loss: As shown in (9), the total loss in LIL contains two balance factors λ_1 and λ_2. In order to further illustrate the impacts of the two losses in avoiding catastrophic forgetting, the ablation experiments for the value of the two balance factors are performed on ten-split AID. The results are shown in Table IV.

When λ_1 and λ_2 are both set to 0.1, BWT $= 12.53 \pm 2.16\%$. Due to insufficient restraint, the model produces catastrophic forgetting of old tasks. With the increase in λ_1 and λ_2, BWT gradually decreases. When both λ_1 and λ_2 are set to 5, BWT reaches a minimum value of $1.35 \pm 1.43\%$. This is in line with our initial design. The greater the weight of dual-constraint loss, the more attention is paid to the precision of feature transfer and the ability to retain old knowledge. It can also be seen from Table IV, compared with the increase in λ_2, BWT decreases more with the increase in λ_1. It shows that the restraint effect of distillation loss is higher than that of adversarial loss, which is consistent with the previous experimental results.

Although BWT is relatively low when λ_1 and λ_2 are 5, mACC also drops. This is because the ability of the model to learn new tasks decreases while reducing the forgetting of old tasks. Combined with the experimental results, we set both λ_1 and λ_2 to 1 in other experiments to maintain the balance of learning new tasks and retaining knowledge of old tasks.

5) Ablation Study for the Number of Tasks: In addition to ten-split AID, we divide the AID dataset equally by the number of categories to generate three-split AID, five-split AID, and 15-split AID. Abundant experiments are performed on these datasets to present the incremental learning performance of LIL on different numbers of tasks. The results of ACL w RB and LIL on a diverse number of tasks of AID are shown in Table V.

It is observed that, when the number of tasks is small (three-split AID and five-split AID), mACC of ACL w RB is low, reaching $58.20 \pm 2.33\%$ and $70.34 \pm 2.12\%$, respectively. When the number of tasks increases, mACC of ACL w RB increases to around 85%. It is proven that the average performance on all tasks of ACL w RB varies greatly with different numbers of tasks. On the contrary, our proposed LIL is less sensitive to the number of tasks. The mACC across all tasks of LIL hardly changes when the amount of tasks changes. Besides, as the number of tasks increases, BWT of both ACL w RB and LIL gradually augments, which indicates that the incremental learning models are bound to produce a small part of forgetting within an acceptable range when there are too many tasks. Compared with ACL w RB, LIL shows outstanding performance in the case of a different number of tasks. Thus, LIL is more robust than ACL w RB.

6) Ablation Study for Generalization: To verify the generalization ability of our proposed method, we apply LIL to state-of-the-art methods designed for remote sensing image scene classification. We choose CLNL [74] and bandwise multiscale (BWMS) [73] as the basic models. Specifically, CLNL uses the cross-layer nonlocal operation to model the long-range correlations of the input images to make the network obtain global contextual understanding ability. BWMS decouples the spectral and spatial feature extractions to capture more

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TABLE VI
RESULTS OF ABLATION STUDY FOR GENERALIZATION

Method	$mACC(\%)$	$BWT(\%)$
CLNL-JT	84.29	-
CLNL-FT	55.37±8.54	35.93±7.16
CLNL-LIL	88.46±3.90	2.67±1.48
BWMS-JT	87.09	-
BWMS-FT	61.24±6.56	30.47±8.27
BWMS-LIL	89.22±2.86	2.03±1.71

comprehensive features from the two aspects and introduces multiscale convolution to adapt to categories with different sizes.

We carry out experiments in three scenarios on these two methods: JT, FT, and adding our proposed method (LIL). In each scenario, we carry out experiments on ten-split AID three times. The same as the previous experiments, the dataset is divided randomly according to categories for each experiment. The experimental results are the mean and standard deviation of the three experiments, as shown in Table VI.

It is observed that the accuracy of JT of these two methods is higher than that of JT in Table I, which indicates that the methods designed for remote sensing images, indeed, achieve superior performance on remote sensing datasets. However, it is inevitable that the two models still face catastrophic forgetting when they are applied in incremental learning scenarios, indicating that models are only suitable for situations where all data are available and well-labeled and cannot dynamically adapt to new data. It can be concluded from Table VI that, when applying LIL to the two methods, catastrophic forgetting is effectively alleviated, and the average accuracy on all tasks of the two models with LIL exceeds JT due to the existence of task IDs. The experimental results prove the versatility and generalization of our proposed LIL, which can be effectively applied to different models.

F. Experiments on EuroSAT

To verify the validity of the proposed model, we further carry out experiments on the large-scale dataset, EuroSAT. We divide EuroSAT into two tasks for incremental learning according to categories, called two-split EuroSAT. Both our method and comparison methods are carried out three times, and the task division for each time is random. Table VII shows the experimental results of mACC and BWT.

It is observed that, even in the case of a small number of tasks, FT still produces large catastrophic forgetting. Compared with FT, the regularization-based methods improve the overall accuracy of all tasks, but the performance of the previous tasks still has a large decline, especially EWC and MAS. For the methods based on parameter expansion, the forgetting rates are small, but the average performance of tasks is still lower than JT. Besides, different from the performance on ten-split AID, ACL faces relatively large forgetting on two-split EuroSAT, and the mACC achieved by ACL w RB is 1.75% lower than that of JT. However, our proposed LIL further improves the average performance of all tasks and makes the mACC 3.29% better than JT. It is concluded that, for the EuroSAT dataset, in the case of a small number of tasks, ACL still produces relatively large forgetting, while the performance of LIL in the previous task drops very little, which proves the effectiveness of our method on the large-scale dataset.

TABLE VII
INCREMENTAL LEARNING RESULTS ON TWO-SPLIT EUROSAT

Method	$mACC(\%)$	$BWT(\%)$
JT	91.56	-
FT	76.44±5.55	15.49±12.05
EWC [20]	79.28±4.69	18.53±4.94
MAS [22]	81.17±5.33	15.05±5.03
LWF [16]	84.34±1.68	10.71±4.71
End-to-End* [18]	84.92±1.74	8.69±4.21
Ammour *et al.* [29]	85.35±1.83	6.75±4.63
PNN [42]	86.52±4.67	-
PathNet [43]	85.08±8.01	3.89±7.78
ACL wo RB [44]	87.75±5.56	4.09±6.39
ACL w RB* [44]	89.81±2.81	3.17±2.51
LIL (ours)	**94.85±0.28**	**0.04±0.07**

TABLE VIII
INCREMENTAL LEARNING RESULTS ON THE PROCESSED BIGEARTHNET

Method	$mACC(\%)$	$BWT(\%)$
JT	92.33	-
FT	62.43±13.18	34.51±13.01
EWC [20]	66.71±6.72	25.19±8.84
MAS [22]	69.39±6.56	21.33±10.19
LWF [16]	74.25±7.79	16.53±7.15
End-to-End* [18]	74.31±5.93	17.21±5.29
Ammour *et al.* [29]	75.35±5.15	15.49±4.36
PNN [42]	79.98±4.19	-
PathNet [43]	81.24±3.56	10.18±2.94
ACL wo RB [44]	85.47±4.37	5.65±2.12
ACL w RB* [44]	88.73±2.19	2.97±1.26
LIL (ours)	**95.89±1.12**	**0.73±0.45**

G. Experiments on BigEarthNet

Furthermore, we verify the effectiveness of our method on another larger scale dataset BigEarthNet. In order to adapt to our experimental settings, we performed a series of preprocessing on BigEarthNet. We first removed 70 987 images covered by seasonal snow, cloud, and cloud shadow based on the officially provided image list, which left 519 339 images. Second, since multilabel classification is not in the scope of

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TABLE IX
RESULTS OF ABLATION STUDY FOR MULTIPLE DATASETS

Method	$mACC(\%)$	$BWT(\%)$
ACL w RB*	70.47±7.08	8.14±5.26
LIL (ours)	**78.61±2.44**	**2.97±2.32**

our research, we selected the images labeled with one category, which contains 103 150 images in total. Third, we chose categories with more than 100 images from 43 categories. There were 102 563 image patches covering 15 categories that were selected. The selected categories contained agroforestry areas, broad-leaved forest, coastal lagoons, complex cultivation patterns, coniferous forest, continuous urban fabric, estuaries, mixed forest, nonirrigated arable land, pastures, peatbogs, permanently irrigated land, sea and ocean, transitional woodland/shrub, and water bodies. Finally, because the images in BigEarthNet have 12 different spectral bands, we selected three RGB spectral bands to composite RGB images for experiments.

For incremental learning experiments, we divided the processed images of 15 categories into five groups, and each group contained three categories. Same as the previous experiment setup, we carried out incremental learning experiments of each method three times and randomly divided the data in each experiment. The experimental results are shown in Table VIII.

Compared with the EuroSAT dataset, the methods based on regularization produce larger forgetting due to the increase in the number of tasks from 2 to 5. The average performance on all tasks of regularization-based methods drops by more than 16% compared with JT. Similar to the previous results, the performance of parameter expansion-based methods is better than that of regularization-based methods, but the superior method ACL w RB only achieves the performance of mACC = 88.73 ± 2.19%, which is 3.6% lower than JT. In all experimental methods, the performance of the proposed LIL reaches the best, achieving mACC = 95.89 ± 1.12% and BWT = 0.73 ± 0.45%, and the average accuracy of all tasks exceeds JT by 3.56%.

Through experiments on large-scale datasets, we once again demonstrate the effectiveness and robustness of LIL. In general, for incremental learning on large datasets, the methods based on regularization still face severe forgetting for the previously learned tasks, and the methods based on parameter expansion are able to alleviate catastrophic forgetting, but the average performance of all tasks is still lower than JT. Even ACL w RB, which is superior to JT on AID, performs poorly on the two large remote sensing image scene classification datasets. On the contrary, our method outperforms JT in both AID and the two large-scale datasets. It shows that our method can be applied to a variety of incremental learning scenarios and proves the generalization ability and robustness of LIL.

H. Experiments on Multiple Datasets

In order to further observe the performance of LIL in a more complex incremental learning scenario, experiments of LIL and ACL w RB are conducted on a sequence of seven remote sensing image scene classification datasets, including AID, EuroSAT, BigEarthNet, RS_C11, RSSCN7, SIRI-WHU, and SAT-4.

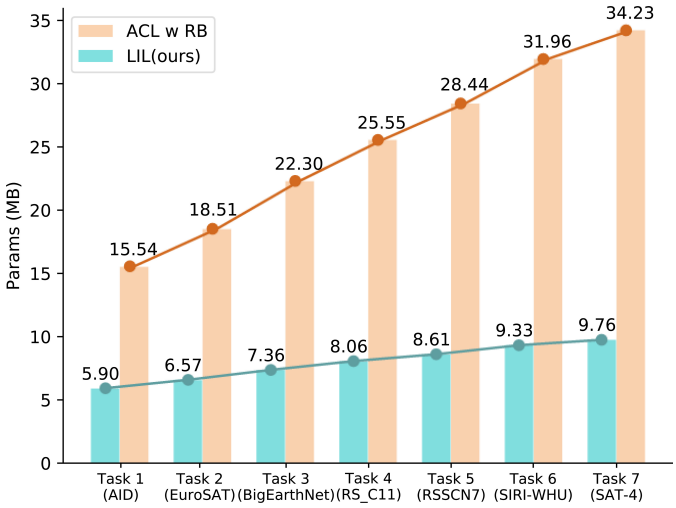


Fig. 10. Number of parameters after learning each task of multiple datasets.

There are differences in capture sensors, shooting areas, data sources, pixel resolutions, and scene categories between disparate datasets. In the experimental settings, each dataset represents one task. Each method performs experiments three times, and the order of learning tasks at each time is randomly selected. The mean and standard deviation of experiments are shown in Table IX. Fig. 10 shows the number of parameters in the models of ACL w RB and LIL after learning each task of seven datasets.

In the experiments, ACL w RB stores five samples for each category of each task, while LIL does not store the data of old tasks. When ACL w RB occupies 34.23-MB parameters, the performances of mACC = 70.47 ± 7.08% and BWT = 8.14 ± 5.26% are achieved. Nevertheless, LIL achieves the excellent performance of mACC = 78.61 ± 2.44% and BWT = 2.97 ± 2.32% with only 9.76-MB parameters. As shown in Fig. 10, the TCE and the FTM proposed in LIL make the initially constructed model in LIL more lightweight than ACL w RB, and the model of LIL also expands fewer parameters than ACL w RB with the increase in the number of learning tasks.

Fig. 11 qualitatively visualizes the experimental results of LIL and ACL w RB on multiple datasets. It can be seen that, for images of similar categories that obey similar distributions, the two methods hardly produce catastrophic forgetting. When images of similar categories obey different distributions, ACL w RB forgets the previous tasks, while the proposed LIL can still perform well on the previous tasks without forgetting. Furthermore, ACL w RB faces serious forgetting when images of different tasks obey greatly different distributions. However, LIL only produces slight forgetting. Due to the large gap between the data, the features are not transferred accurately to the original feature space, and the classification modules are not updated on the new transferred features, resulting in

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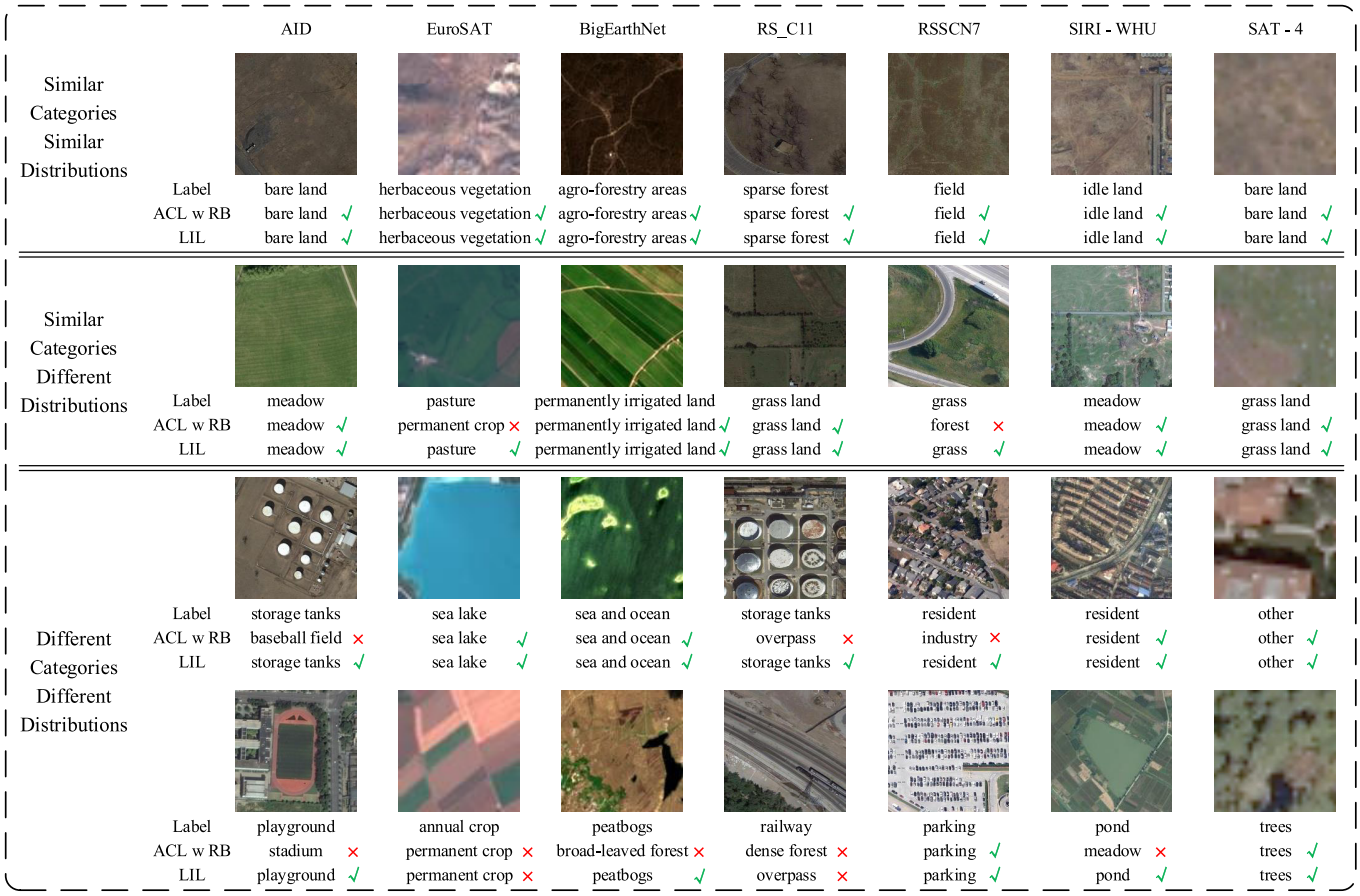


Fig. 11. Qualitative comparisons between LIL and ACL w RB on multiple datasets.

slight forgetting on previous tasks. Even with slight forgetting, our method still achieves a performance that exceeds ACL w RB in complex scenarios.

Effect of the Order of Learning Tasks: In previous experiments, the order of learning tasks in each experiment was randomly selected. The FTM proposed in LIL maps features between the feature spaces of adjacent tasks, so there may be an impact on the performance of the model caused by the order of learning tasks. However, it can be seen from previous results that the standard deviation of the three results of LIL is smaller than other compared methods, which indicates that the order of learning task has less effect on our method.

To further verify the influence of the order of learning tasks on our model, we artificially specified two task orders for experiments on seven datasets. Fig. 12 shows the data distributions of the seven datasets after dimensionality reduction using t-SNE. Among them, Fig. 12(a) represents a designed order in which the images of adjacent learning tasks are relatively similar. In the other designed sequence, the data of adjacent tasks are quite different, as shown in Fig. 12(d). In addition, as shown in Fig. 12(b) and (c), we also represent two random orders of tasks. Table X shows the experimental results of the four orders of learning tasks.

As expected, the result of order (a) is the best among the four orders, reaching mACC = 79.86% and BWT = 1.05%. The result of order (d) is the worst, but it is still better than that of the state-of-the-art incremental learning method ACL w RB.

TABLE X
INCREMENTAL LEARNING RESULTS OF THE ORDER OF LEARNING TASKS ON MULTIPLE DATASETS

Learning order	$mACC(\%)$	$BWT(\%)$
(a)	79.86	1.05
(b)	77.78	3.12
(c)	78.93	2.69
(d)	77.19	3.87

The average accuracy of all tasks is 6.72% higher than ACL w RB, and the forgetting rate is 4.27% lower than ACL w RB. On the contrary, compared with order (a), the average performance of all tasks merely drops by 2.67%, and the forgetting rate only increases by 2.82%. This demonstrates that LIL achieves state-of-the-art performance even with large differences between the data of adjacent tasks. Besides, the results of the remaining two randomly selected orders (b) and (c) are between the performance of order (a) and (d).

The experimental results show that, if the difference of the data between adjacent tasks is small, the FTM is able to map features of the adjacent tasks effectively, and the performance of our model is preeminent. Even if the data between adjacent tasks are extremely different, the performance of our method is only slightly degraded, and it still surpasses the current

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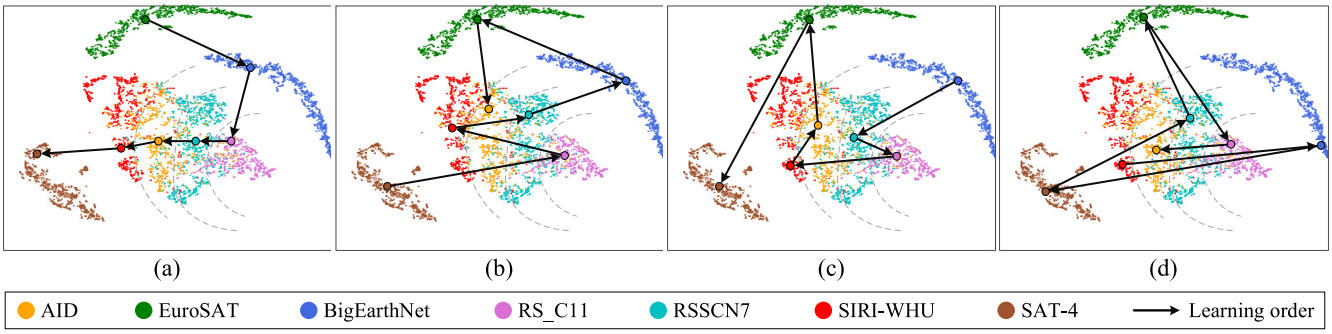


Fig. 12. Order of learning tasks on multiple datasets. (a) Data of adjacent tasks in the order is relatively similar. (b) and (c) Randomly selected order of learning tasks. (d) There are large differences between the data of adjacent learning tasks.

advanced method ACL w RB. Thus, it can be concluded that the order of learning tasks has an impact on our model, but only slightly. Even in the scenario where the data of different tasks have a large difference, LIL still achieves the state-of-the-art performance of all tasks.

In consequence, it is encouraging to see that our proposed LIL is equipped to maintain splendid performance in the complex situations of incremental learning with a lightweight architecture. It is not only able to consecutively learn on a single dataset but also across various datasets.

V. Conclusion

In this article, we propose a novel LIL method in the field of remote sensing image scene classification, named LIL. It only adds a light FTM and a classifier to keep the performance on old tasks while learning the new task. We replace the mission of the feature extractor to extract features in the feature space of the newest learned task, which is able to lighten the backbone. A tiny FTM is proposed to implement the transfer and alignment of data distributions between neighboring tasks in the feature spaces. In addition, dual-constraint loss with knowledge distillation and adversarial training is designed to supervise the training of the proposed FTM at both feature level and semantic level. Without saving the data of old tasks, LIL alleviates catastrophic forgetting with a small incremental amount of parameters. Extensive experiments on a single dataset and a sequence of multiple datasets reveal the effectiveness and efficiency of our proposed method for incremental learning in remote sensing image scene classification.

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